Quizrr Chapter-Wise Test for JEE Advanced - 2025

By: C I P H Ξ R

Q.1

At the end points A, B of the fixed segment of length L, lines are drawn meeting in C and making angles θ and 2θ respectively with the given segment. Let D be the foot of the altitude CD and let x represents the length of AD. The value of x as θ tends to zero i.e. $\lim_{\theta \to 0} x$ equals -

(A) L/2

(B) $2\,\mathrm{L}/3$

(C) 3L/4

(D) L/4

Q.2

Let $\mathrm{f}(\mathrm{x})$ is a function continuous for all $\mathrm{x} \in \mathrm{R}$ except at $\mathrm{x} = 0$ such that $\mathrm{f}'(\mathrm{x}) < 0\ \forall \mathrm{x} \in (-\infty, 0)$ and $\mathrm{f}'(\mathrm{x}) > 0 \quad \forall \mathrm{x} \in (0, \infty)$. If $\lim_{\mathrm{x}\to 0^+} \mathrm{f}(\mathrm{x}) = 3, \lim_{\mathrm{x}\to 0^-} \mathrm{f}(\mathrm{x}) = 4$ and $\mathrm{f}(0) = 5$, then the image of the point $(0, 1)$ about the line $y \lim_{x\to 0} f\left(\cos^3 x - \cos^2 x\right) = x \lim_{x\to 0} f\left(\sin^2 x - \sin^3 x\right)$ is

(A) $\left(\frac{12}{25}, \frac{-9}{25}\right)$

(B) $\left(\frac{12}{25}, \frac{9}{25}\right)$

(C) $\left(\frac{16}{25}, \frac{-8}{25}\right)$

(D) $\left(\frac{24}{25}, \frac{-7}{25}\right)$

Q.3

Let $\mathrm{a} = \min\left[\mathrm{x}^2 + 2\mathrm{x} + 3\right], \quad \mathrm{x} \in \mathrm{R}$ and $\mathrm{b} = \mathrm{Lim}_{\mathrm{x}\to 0} \frac{\sin \mathrm{x} \cos \mathrm{x}}{\mathrm{e}^{\mathrm{x}} - \mathrm{e}^{-\mathrm{x}}}$. Then the value of $\sum_{\mathrm{r}=0}^{\mathrm{n}} \mathrm{a}^{\mathrm{r}} \mathrm{b}^{\mathrm{n}-\mathrm{r}}$ is $-$

(A) $\frac{2^{\mathrm{n}+1}+1}{3 \cdot 2^{\mathrm{n}}}$

(B) $\frac{2^{n+1}-1}{3 \cdot 2^{n}}$

(C) $\frac{2^{\mathrm{n}}-1}{3 \cdot 2^{\mathrm{n}}}$

(D) $\frac{4^{\mathrm{n}+1}-1}{3 \cdot 2^{\mathrm{n}}}$

Q.4

If x is a real number in $[0, 1]$ then the value of $\underset{m\to\infty}{\text{Limit}}\ \text{Limit}_{n\to\infty} \left[1 + \cos^{2m}(n!\pi x)\right]$ is given by

(A) 1 or 2 according as x is rational or irrational

(B) 2 or 1 according as x is rational or irrational

(C) 1 for all x

(D) 2 for all x.

Q.5

If P denotes the number of point of intersection of $\mathrm{y}^2 = 7\mathrm{x}$ and $\mathrm{x}^2 + \mathrm{y}^2 - 4\mathrm{x} + 2 = 0$, then value of $\lim\limits_{\mathrm{x}\to \mathrm{p}} \frac{\mathrm{x}\sin(\sin\ x) - \sin^2 \mathrm{x}}{\mathrm{x}^6}$, is-

(A) $\frac{1}{6}$

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(B) $\frac{1}{18}$

(C) $\frac{1}{12}$

(D) $\frac{1}{24}$

Q.6

If $f(x)=\begin{cases}\left(p^2-1\right)(\{x\}+2[x])-2, & -2<x\leq -1\\ q\left(\frac{e^x+e^{-x}}{2}\right)+|p|(x-1), & -1<x<2\end{cases}$ $,p,q\in R$ is continuous in $(-2,2)$ then $f\left(f\left(f\left(\frac{-1}{2}\right)\right)\right)$ is

(A) -2

(B) -1

(C) 0

(D) not defined

[Note: [k] denotes greatest integer function less than or equal to k and $\{k\}$ denotes fractional part function of k.]

Q.7

Given

$f(x)=\begin{cases}x^2e^{2(x-1)} \text{ for } 0\leq x\leq 1 & ;0\leq x\leq 1\\ a\,\mathrm{sgn}(x+1)\cos(2x-2)+bx^2; & 1<x\leq 2\end{cases}$

is differentiable at $x=1$. Then :

(A) $a=-1,b=2$

(B) $a=1,b=-2$

(C) $a=-3,b=4$

(D) None of these

Q.8

$f(x)=\begin{cases}\max.\{e^x,e^{-x},2\}\,x\leq 0\\ \min,\{e^x,e^{-x},2\}\,x>0\end{cases}$ which of the following statements is not true?

(A) $f(x)$ is discontinuous at $\mathrm{x}=0$

(B) $f(x)$ is non-derivable at exactly two points.

(C) $f(x)$ has non removable type of discontinuity at $\mathrm{x}=0$ with jump of discontinuity equal to 2 .

(D) $f(x)$ is continuous but non-derivable at $x=\log_{\epsilon}\frac{1}{2}$

Q.9

Consider $f(x)=\frac{1}{\ln|x|}$. Which of the following alternative (s) is/are correct-

(A) Number of points of discontinuity in function is 3.

(B) $\lim_{x\to 0}f(x)$ is not existent

(C) Number of inflection point(s) on the graph is 2 .

(D) $\mathrm{f}(\mathrm{x})$ has two asymptotes

Q.10

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$\lim_{x\to 1} \frac{(1-x)(1-x^2)(1-x^3).....(1-x^{2n})}{[(1-x)(1-x^2)(1-x^3).....(1-x^n)]^2}$ is equal to -

(A) $\prod_{r=1}^{n} \frac{n+r}{r}$

(B) $\frac{1}{n!}\prod_{r=1}^{n}(4r-2)$

(C) The sum of the coefficients of two middle terms in the expansion of $(1+x)^{2n-1}$

(D) The coefficient of x^n in the expansion of $(1+x)^{2n}$.

Q.11

Which of the following function(s) has/have removable discontinuity at $x = 1$.

(A) $f(x) = \frac{1}{\ln|x|}$

(B) $f(x) = \frac{x^2-1}{x^3-1}$

(C) $f(x) = 2^{-2^{\frac{1}{1-x}}}$

(D) $f(x) = \frac{\sqrt{x+1}-\sqrt{2x}}{x^2-x}$

Q.12

Passage

For $0 < x < \frac{\pi}{2}$, let $f_1(x) = \sum_{r=1}^{n} \sec\left(x + \frac{r\pi}{6}\right)\sec\left(x + (r-1)\frac{\pi}{6}\right)$ and $f_1(x) - 2f_2(x) = 2\tan\left(x + \frac{n\pi}{6}\right)$.

$$\text{Also, } f_2(x) + f_3(x) = 0 \text{ and } f_4(x) = \begin{cases} \dfrac{e^{(e^x + f_2(x) + \tan x}-1}{e^{2(e^x-1)}} & ; x < 0 \\ k_1 & ; x = 0 \\ (1+|f_2(x)|)^{\frac{k_2}{f_3(x)}} & ; x > 0 \end{cases}$$

Question:

The value of k_1 and k_2, if $f_4(x)$ is continuous at $x = 0$ is -

(A) $\frac{1}{\sqrt{e}}, 2$

(B) e, 1

(C) $\frac{e}{2}, 2$

(D) $\sqrt{e}, \frac{1}{2}$

Q.13

Passage

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For $0 < x < \frac{\pi}{2}$, let $f_1(x) = \sum_{r=1}^{n} \sec\left(x + \frac{r\pi}{6}\right) \sec\left(x + (r-1)\frac{\pi}{6}\right)$ and $f_1(x) - 2f_2(x) = 2\tan\left(x + \frac{n\pi}{6}\right)$

$$\text{Also, } f_2(x) + f_3(x) = 0 \text{ and } f_4(x) = \begin{cases} \dfrac{e^{(e^{x+f_2(x)+\tan x}} - 1}{e^{2(e^x-1)}} & ; x < 0 \\ k_1 & ; x = 0 \\ (1+|f_2(x)|)^{\frac{k_2}{f_3(x)}} & ; x > 0 \end{cases}$$

Question:

$y = f_3(x)$ is –

(A) discontinuous and non-derivable at $x = \pi/4$ and $\pi/3$

(B) neither continuous nor derivable at $x = 2\pi/5$

(C) continuous and derivable in $(0, \pi/2)$

(D) continuous but not derivable at $x = 2\pi/5$

Q.14

Passage

For $0 < x < \frac{\pi}{2}$, let $f_1(x) = \sum_{r=1}^{n} \sec\left(x + \frac{r\pi}{6}\right) \sec\left(x + (r-1)\frac{\pi}{6}\right)$ and $f_1(x) - 2f_2(x) = 2\tan\left(x + \frac{n\pi}{6}\right)$

$$\text{Also, } f_2(x) + f_3(x) = 0 \text{ and } f_4(x) = \begin{cases} \dfrac{e^{(e^{x+f_2(x)+\tan x}} - 1}{e^{2(e^x-1)}} & ; x < 0 \\ k_1 & ; x = 0 \\ (1+|f_2(x)|)^{\frac{k_2}{f_3(x)}} & ; x > 0 \end{cases}$$

Question:

For $n = 3$, the solution of equation $f_1(x) + 4 = 0$, in $(0, \pi/2)$ is

(A) $\pi/6$

(B) $\pi/4$

(C) $\pi/3$

(D) non existent

Q.15

Passage

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The function $f(x)=\begin{cases} ax(x-1)+b; & x<1 \\ x-1; & 1 \leqslant x \leqslant 3 \\ px^2+qx+2; & x>3 \end{cases}$
such that (1) $f(x)$ is continuous $\forall x \in R$,
(2) $f'(1)$ does not exist,
(3) $f'(x)$ is continuous at $x=3$
Question: Value of b is -

(A) 0

(B) $1/3$

(C) -1

(D) 1

Q.16

Passage

The function $f(x)=\begin{cases} ax(x-1)+b; & x<1 \\ x-1; & 1 \leqslant x \leqslant 3 \\ px^2+qx+2; & x>3 \end{cases}$
such that (1) $f(x)$ is continuous $\forall x \in R$,
(2) $f'(1)$ does not exist,
(3) $f'(x)$ is continuous at $x=3$
Question:
Value of p is -

(A) 0

(B) $1/3$

(C) -1

(D) 1

Q.17

Passage

The function $f(x)=\begin{cases} ax(x-1)+b; & x<1 \\ x-1; & 1 \leqslant x \leqslant 3 \\ px^2+qx+2; & x>3 \end{cases}$
such that (1) $f(x)$ is continuous $\forall x \in R$,
(2) $f'(1)$ does not exist,
(3) $f'(x)$ is continuous at $x=3$
Question:
Value of q is -

(A) 0

(B) $1/3$

(C) -1

(D) 1

Q.18

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Match the column -

(A) If $f(x) = \begin{cases} \dfrac{\left(\frac{\pi}{2} - \sin^{-1}(1-\{x\}^2)\right)\sin^{-1}(1-\{x\})}{\sqrt{2}\,(\{x\}-\{x\}^3)}, & x>0 \\ k, & x=0 \\ \dfrac{A\sin^{-1}(1-\{x\})\cos^{-1}(1-\{x\})}{\sqrt{2\{x\}}\,(1-\{x\})}, & x<0 \end{cases}$ (p) 2

is continuous at $x = 0$, then the value of $\sin^2 k + \cos^2\left(\frac{A\pi}{\sqrt{2}}\right)$, is

(where $\{x\}$ denotes fractional part function)

(B) If $f(x) = [2 + 5 \mid n \mid \sin x]$ where $n \in I$ has exactly 19 points of non-derivability in $(0, \pi)$, then possible value(s) of n are (where $[x]$ denotes greatest integer function) (q) – 2

(C) Let $f(x) = \begin{cases} x\dfrac{\left(\frac{3}{4}\right)^{1/x} - \left(\frac{3}{4}\right)^{-1/x}}{\left(\frac{3}{4}\right)^{1/x} + \left(\frac{3}{4}\right)^{-1/x}}, & x \neq 0 \\ 0, & x = 0 \end{cases}$ (r) 3

If $P = f'(0^-) - f'(0^+)$, then $\lim\limits_{x \to P^-} \dfrac{(\exp((x+2)\ln 4))^{\frac{[x+1]}{4}} - 16}{4^x - 16}$ (s) –3

is less than (where $[x]$ denotes greatest integer function)

(A) A-p ; B-p,q ; C-p,r

(B) A-p ; B-q,r ; C-p

(C) A-q,r ; B-p,q ; C-r,q

(D) A-p,q ; B-q ; C-p,q,r

Q.19

For positive integers $k = 1, 2, 3, \ldots..n$, let S_k denotes the area of ΔAOB_k (where O is origin) such that $\angle AOB_k = \frac{k\pi}{2n}$, $OA = 1$ and $OB_K = k$. The value of the $\lim_{n\to\infty} \frac{1}{n^2} \sum_{k=1}^{n} S_k$ is $\frac{X}{\pi^2}$. Find the value of X.

Q.20

If $\lim_{x\to\infty} \frac{a(2x^3-x^2)+b(x^3+5x^2-1)-c(3x^3+x^2)}{a(5x^4-x)-bx^4+c(4x^4+1)+2x^2+5x} = 1$, then the value of $(a+b+c)$ can be expressed in the lowest form as $\frac{p}{q}$. The value of $(p+q-160)$ is.

Answers & Solutions

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Q.1 Answer:

$2\,L/3$

Solution:

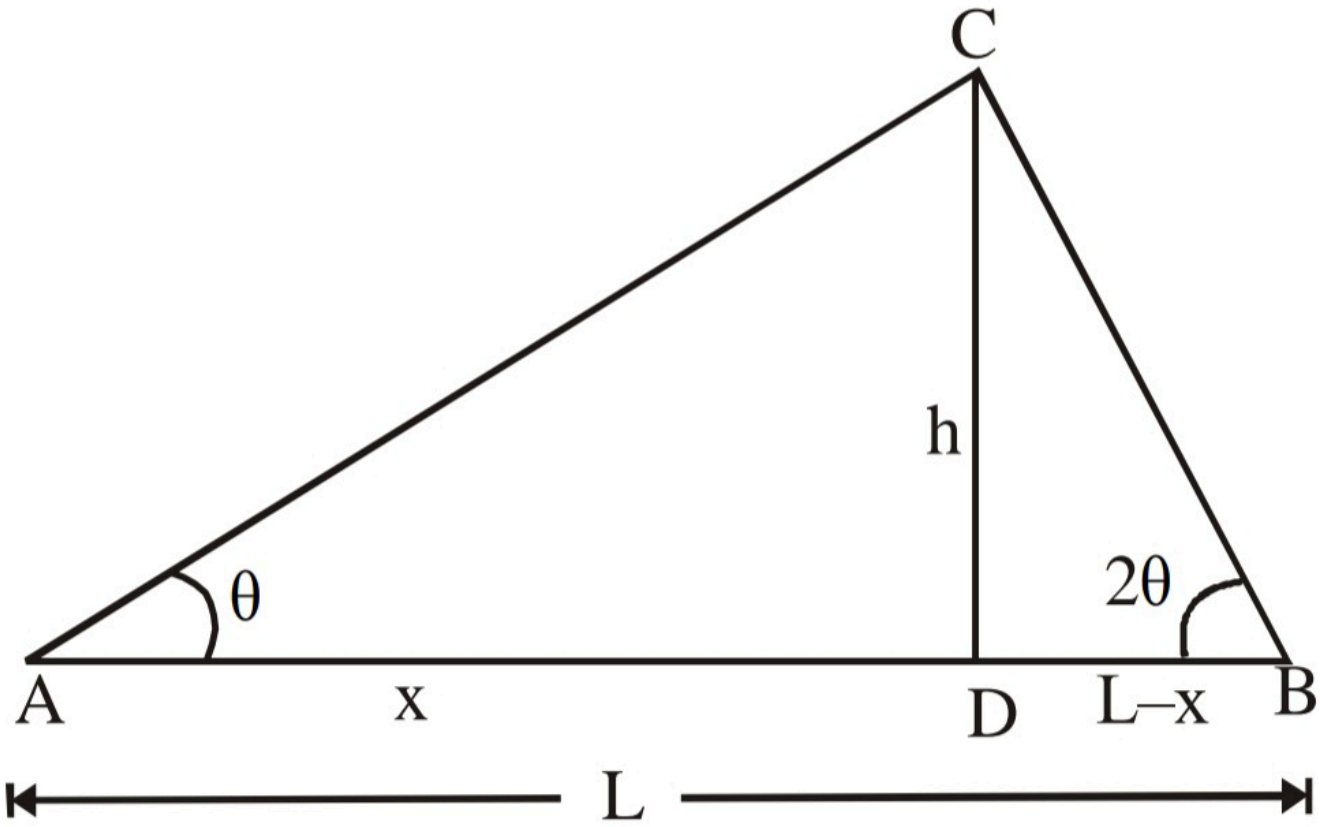
$\mathrm{CD} = \mathrm{h}, \tan\theta = \frac{\mathrm{h}}{\mathrm{x}}, \tan 2\theta = \frac{\mathrm{h}}{\mathrm{L}-\mathrm{x}}$

Now, $x\tan\theta = (L-x)\tan 2\theta$

$x(\tan\theta + \tan 2\theta) = L\tan 2\theta$

$\mathrm{x} = \left(\frac{\tan 2\theta}{\tan\theta + \tan 2\theta}\right)\mathrm{L}$

$\mathrm{x} = \left(\frac{2\theta\frac{\tan 2\theta}{2\theta}}{\theta\frac{\tan\theta}{\theta} + 2\theta\frac{\tan 2\theta}{2\theta}}\right)\mathrm{L}$



$\lim_{\theta\to 0}\mathrm{x} = \frac{2\,\mathrm{L}}{3}$

Q.2 Answer:

$\left(\frac{24}{25}, \frac{-7}{25}\right)$

Solution:

We have $y\lim_{x\to 0} f\left(\cos^3 x - \cos^2 x\right) = x\lim_{x\to 0} f\left(\sin^2 x - \sin^3 x\right)$

$\cos^3 x - \cos^2 x \to 0$ **from L.H.S.**

$\therefore \lim_{x\to 0} f\left(\cos^3 x - \cos^2 x\right) = 4$

$\sin^2 x - \sin^3 x \to 0$ from R.H.S.

$\therefore \lim_{x\to 0} f\left(\sin^2 x - \sin^3 x\right) = 3$

$\therefore$ Equation of the line is $4y = 3x$ ..... (1)

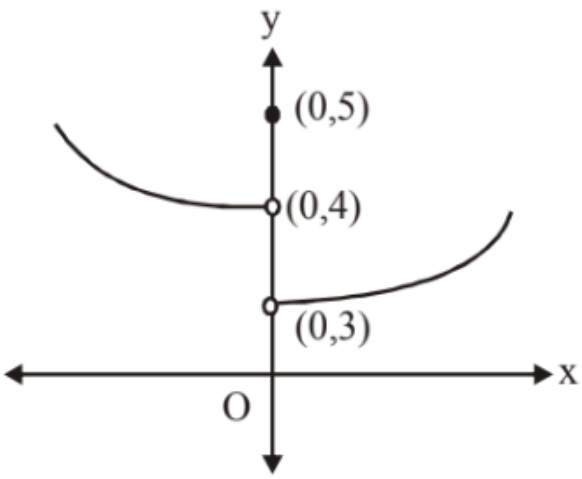
Equation of line perpendicular to $y = \frac{3}{4}x$ and

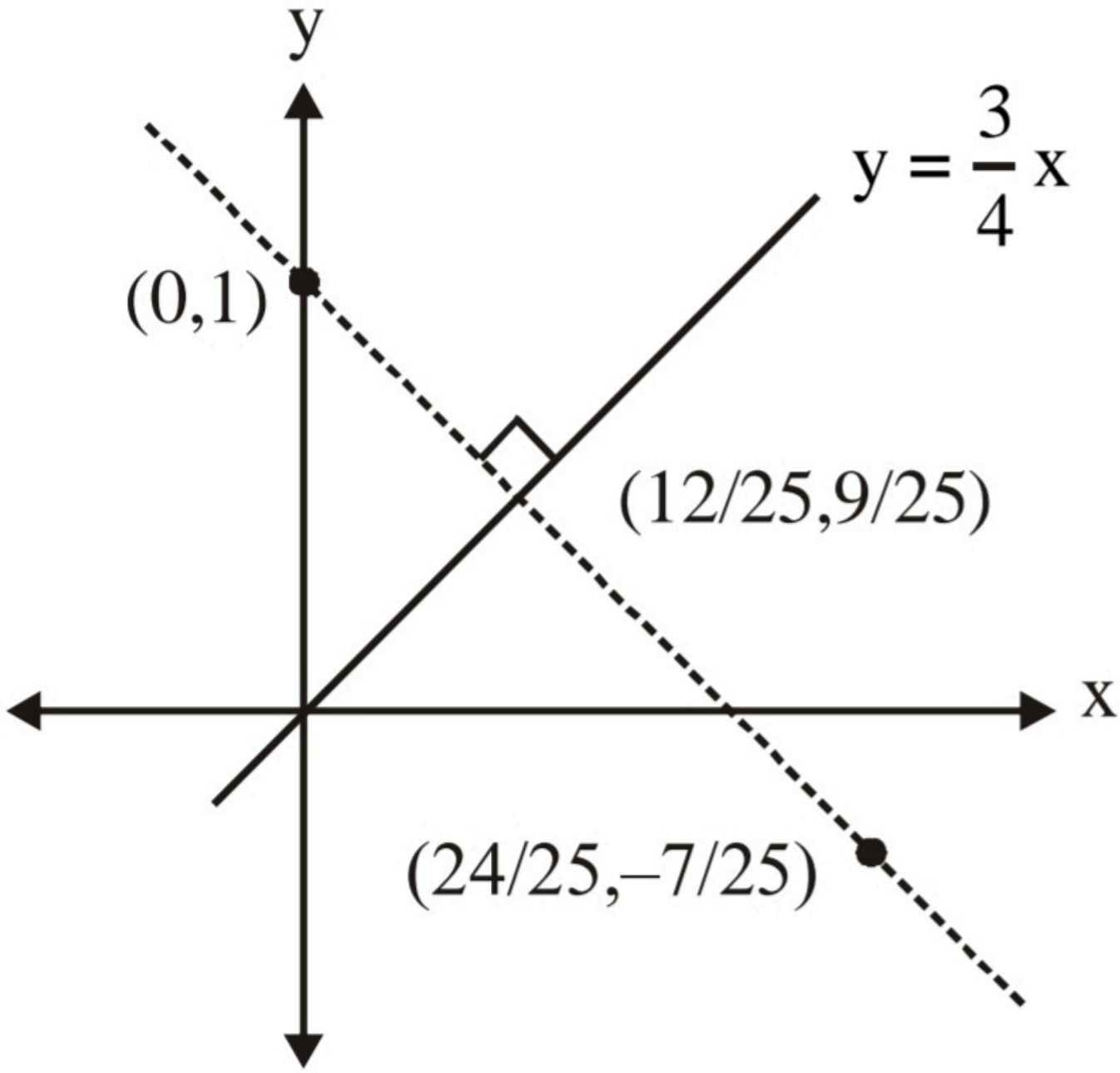
$y - 1 = \frac{-4}{3}x$

On solving eq. (1) and eq. (2), we get $x = \frac{12}{25}, y = \frac{9}{25}$

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Hence image point is $\left(\frac{24}{25}, \frac{-7}{25}\right)$





Q.3 Answer:

$\frac{4^{n+1}-1}{3 \cdot 2^n}$

Solution:

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$a = (x+1)^2 + 2 \Rightarrow a = 2$

$\mathrm{b} = \mathrm{Lim}_{\mathrm{x}\to 0} \frac{\sin 2\mathrm{x}}{\frac{2(e^{2\mathrm{x}}-1)\cdot 2\mathrm{x}}{2\mathrm{x}}} = \frac{1}{2}.$

Now $\sum_{\mathrm{r}=0}^{\mathrm{n}} \mathrm{a}^{\mathrm{r}}\mathrm{b}^{\mathrm{n}-\mathrm{r}} = \sum 2^{\mathrm{r}} \left(\frac{1}{2}\right)^{\mathrm{n}-\mathrm{r}} = \frac{1}{2^{\mathrm{n}}} \sum_{\mathrm{r}=0}^{\mathrm{n}} 2^{2\mathrm{r}}$

$= \frac{1}{2^n} \sum_{r=0}^{n} 4^r = \frac{1}{2^n} \times \left[1 + 4 + 4^2 + \ldots.. + 4^n\right] = \frac{1}{2^n}\left[\frac{4^{n+1}-1}{3}\right] = \frac{4^{n+1}-1}{3.2^n}$

Q.4 Answer:

2 or 1 according as x is rational or irrational

Solution:

$\underset{m\to\infty}{\mathrm{Limit}}\, \mathrm{Limit}_{n\to\infty}\, 1 + \cos^2\ \mathrm{m}(\mathrm{n}!\pi\mathrm{n})$

Let $x \in [0,1]$ **is a rational**

then $\mathrm{n}!.\mathrm{x}$ **if** $\mathrm{n} \to \infty$ **becomes integral**

$\therefore \quad \cos^2(\mathrm{n}!\mathrm{x})\pi = 1$

$\therefore \quad \mathrm{Limit}_{m\to\infty} \left(\cos^2 n!x\pi\right)^m = 1$

$\therefore \quad l = 1 + 1 = 2$ **if** x **is rational**

if x **is irrational, then** $n!x$ **is not an integer**

$\therefore \quad \cos^2(\mathrm{n}!\mathrm{x})\pi$ **is less than 1**

$(\cos^2(n!x)\pi)^m \to 0$

$\therefore \quad \mathrm{l} = 1 + 0 = 1 \Rightarrow$ **the result**

Q.5 Answer:

$\frac{1}{18}$

Solution:

$L = \lim_{x\to 0} \frac{\sin^2 x}{x^2}\left[\frac{x\sin(\sin x)}{\sin^2 x} - 1\right]\frac{1}{x^4}$

$= \lim_{x\to 0} \frac{1}{x^4}\frac{\sin^2 x}{x^2}\left[\frac{x}{\sin x}\left\{1 - \frac{\sin^2 x}{6} + \frac{\sin^4 x}{120} - 1\right\}\right]$

$\lim_{x\to 0} \frac{1}{x^4} \frac{\left(1 - \frac{\sin^2 x}{6} + \frac{\sin^4 x}{120}\ldots\left(1 - \frac{x^2}{6} + \frac{x^4}{120}\ldots\right)\right)}{1 - \frac{x^2}{6} + \ldots}$

$= \lim_{x\to 0} \frac{1}{x^4}\left[-\frac{x^2}{6}\left(1 - \frac{x^2}{6} + \ldots\right)^2 + \frac{x^4}{120} + \frac{x^6}{6} - \frac{x^4}{120} + ..\right]$

$= \lim_{x\to 0} \frac{1}{x^4}\left[\frac{x^4}{18} + \ldots\right] = \frac{1}{18}$

Q.6 Answer:

not defined

[Note: [k] denotes greatest integer function less than or equal to k and $\{k\}$ denotes fractional part function of k.]

Solution:

$$f(x) = \begin{cases} \left(p^2-1\right)(x-2) - 2; & -2 < x < -1 \\ -2\left(p^2-1\right) - 2; & x = -1 \\ q\left(\frac{e^x + e^{-x}}{2}\right) + |p|(x-1); & -1 < x < 2 \end{cases}$$

$$-3\left(p^2-1\right) - 2 = -2\left(p^2-1\right) - 2 = q\left(\frac{e^{-1}+e^1}{2}\right) - 2|p| \Rightarrow p = \pm 1, q = 0$$

$$f(x) = \begin{cases} -2; & -2 < x < -1 \\ -2; & x = -1 \\ x - 1; & -1 < x < 2 \end{cases}$$

$f\left(f\left(f\left(\frac{-1}{2}\right)\right)\right) = f\left(f\left(\frac{-3}{2}\right)\right) = f(-2) =$ **Not defined**

Q.7 Answer:

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$a = -1, b = 2$

Solution:

L.H.L. $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} x^2 e^{2(x-1)} = 1$

Also, $f(1) = 1$

Now R.H.L. $= \lim_{x\to 1^-} f(x) = \lim_{x\to 1^+} a\,\text{sgn}(x+1)\cos 2$

$(x-1) + bx^2 = a \cdot 1 \cdot 1 + b$

For continuity $a + b = 1$

Also, LHD $(x = 1)$ **is**

$\lim_{h\to 0} \frac{(1-h)^2 e^{-2h} - 1}{h} = \lim_{h\to 0} 2e^{-2h} + he^{-2h}\left(\frac{e^{-2h}-1}{h}\right)$

$= 2 + 0 + 2 = 4$

RHD $(x = 1)$ **is** $\lim_{h\to 0} \frac{a\,\text{sgn}(2+h)\cos 2h + b(1+h)^2 - 1}{h}$

$= \lim_{h\to 0} \frac{a\cos 2h + b + bh^2 + 2bh - (a+b)}{h}$

$= \lim_{h\to 0} \left(\frac{\cos 2h - 1}{h}\right) + bh + 2b = 2b$

$(\because a + b = 1)$

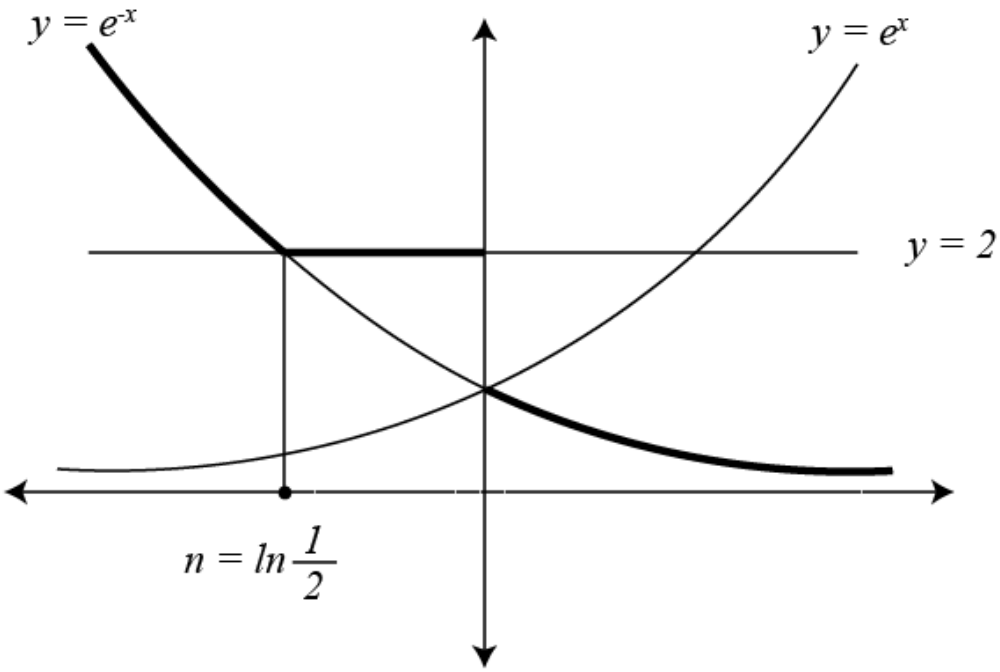
$f(x)$ **is differentiable at** $x = 1$ **if** $2b = 4$

$\Rightarrow b = 2; a = -1$

Q.8 Answer:

$f(x)$ has non removable type of discontinuity at $\mathrm{x} = 0$ with jump of discontinuity equal to 2 .

Solution:



The bold graph is of $f(x)$

(1) $f(x)$ **is discontinues at** $x = 0$

(2) $f(x)$ **is non-derivable at two points** $\left(x = \ln\frac{1}{2}, 0\right)$

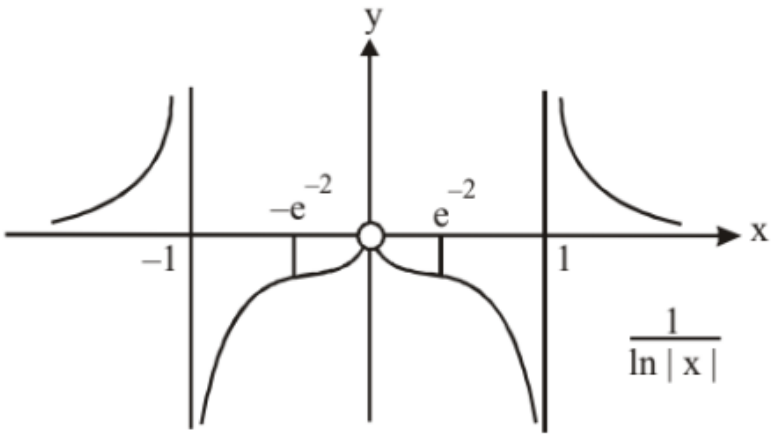
(4) $f(x)$ **is continuous at** $x = \ln\frac{1}{2}$ **but non derivable.**

Q.9 Answer:

Number of points of discontinuity in function is 3.

Solution:

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$f(x) = \frac{1}{\ln|x|} = \begin{cases} \frac{1}{\ln x} \text{ if } x > 0 \\ \frac{1}{\ln(-x)} \text{ if } x < 0 \end{cases}$

Q.10 Answer:

$\prod_{r=1}^{n} \frac{n+r}{r}$

Solution:

Dividing each term of N^r **and** D^r **by**
$(1-x)$ **are using the limit** $\lim_{x\to 1} \frac{1-x^m}{1-x} = m$,
we get $\ell = \frac{1.2.3.4.\ldots..2n}{(1.2.3.4.\ldots n)^2} = \frac{(2n)!}{n!n!} = {}^{2n}C_n$
which is also equal to the coefficient of x^n **in** $(1+x)^{2n} \Rightarrow (D)$ **is correct**
Also ${}^{2n}C_n = \frac{2n!}{n!.n!} = \frac{2^n \cdot n![1.3.5.\ldots..2n-1]}{n!n!} = \frac{2.6.10.\ldots..(4n-2)}{1.2.3.\ldots..n} = \frac{1}{n!}\prod_{r=1}^{n}(4r-2) \Rightarrow$ **(B)**
Again $\frac{2n!}{n!n!} = \frac{(n+1)(n+2).\ldots.(n+n)}{1.2.3.\ldots n} = \prod_{r=1}^{n} \frac{n+r}{r} \Rightarrow$ **(A)**
now coefficient of two middle terms in $(1+x)^{2n-1}$
$=$ **coefficient of** $x^n +$ **coefficient of** $x^{n-1} = 2n-1C_n + {}^{2n-1}C_n = {}^{2n}C_n \Rightarrow$ **(C)**

Q.11 Answer:

$f(x) = \frac{x^2-1}{x^3-1}$

Solution:

(A) $\lim_{x\to 1} f(x)$ **does not exist.**
(B) $\lim_{x\to 1} f(x) = \frac{2}{3}$
$\therefore$ $f(x)$ **has removable discontinuity at** $x = 1$
(C) $\lim_{x\to 1} f(x)$ **does not exist.**
(D) $\lim_{x\to 1} f(x) = \frac{-1}{2\sqrt{2}}$ $\therefore$ $f(x)$ **has removable discontinuity at** $x = 1$

Q.12 Answer:

$\sqrt{e}, \frac{1}{2}$

Solution:

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We have, $f_1(x) = 2\sum_{r=1}^{n} \frac{\sin\left(\left(x+\frac{r\pi}{6}\right)-\left(x+(r-1)\frac{\pi}{6}\right)\right)}{\left(x+(r-1)\frac{\pi}{6}\right)\cdot\cos\left(x+\frac{r\pi}{6}\right)}$

$$= 2\left[\left(\tan\left(x+\frac{\pi}{6}\right)-\tan x\right)+\left(\tan\left(x+\frac{2\pi}{6}\right)-\tan\left(x+\frac{\pi}{6}\right)\right)+\left(\tan\left(x+\frac{3\pi}{6}\right)-\tan\left(x+\frac{2\pi}{6}\right)\right)+....+\left(\tan\left(x+\frac{n\pi}{6}\right)-\tan\left(x+(n-1)\frac{\pi}{6}\right)\right)\right]$$

$$\Rightarrow \quad f_1(x) = 2\left(\tan\left(x+\frac{n\pi}{6}\right)-\tan x\right)$$

$$\text{for } n = 3,\ f_1(x) = 2\left(\tan\left(\frac{\pi}{2}+x\right)-\tan x\right) = 2(-\cot x - \tan x) = -2\frac{1}{\sin x \cos x}$$

$$2f_2(x) = f_1(x) - 2\tan\left(x+\frac{n\pi}{6}\right) = 2\tan\left(x+\frac{n\pi}{6}\right) - 2\tan x - 2\tan\left(x+\frac{n\pi}{6}\right)$$

$$\therefore \quad f_2(x) = -\tan x$$

$$f_3(x) = -f_2(x), \text{ so}$$

$$f_3(x) = \tan x$$

$$\text{Now, } f_4(x) = \begin{cases} \frac{e^{(e^x-1)}-1}{e^{2(e^x-1)}} & ; x < 0 \\ k_1 & ; x = 0 \end{cases}$$

$$\text{Clearly, } f\left(0^-\right) = e^{\frac{1}{2}\ln e} = \sqrt{e}$$

$$f\left(0^-\right) = e^{k_2} \& f(0) = k_1$$

(i) $\text{As } f_4(x)$ **is continuous at** $x = 0$**, so be definition of continuity, we get** $f\left(0^-\right) = f\left(0^+\right) = f(0)$

$$\sqrt{e} = e^{k_2} = k_1 \quad \therefore \quad k_1 = \sqrt{e} \& k_2 = \frac{1}{2}$$

(ii) As $y = f_3(x) = \tan x$

Clearly $f_3(x)$ **is continuous as well as derivable everywhere in** $(0, \pi/2)$

(iii) $n = 3, f_1(x) = -4$

$$\Rightarrow \quad -2(\tan x + \cot x) = -4$$

$$\Rightarrow \quad \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = 2$$

$$\Rightarrow \quad 1 = 2\sin x \cdot \cos x \quad \therefore \quad x = \frac{\pi}{4} \in \left(0, \frac{\pi}{2}\right)$$

Q.13 Answer:

continuous and derivable in $(0, \pi/2)$

Solution:

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We have, $f_1(x) = 2\sum_{r=1}^{n} \frac{\sin\left(\left(x+\frac{r\pi}{6}\right)-\left(x+(r-1)\frac{\pi}{6}\right)\right)}{\cos\left(x+(r-1)\frac{\pi}{6}\right)\cdot\cos\left(x+\frac{r\pi}{6}\right)}$

$$= 2\left[\begin{array}{l}\left(\tan\left(x+\frac{\pi}{6}\right)-\tan x\right)+\left(\tan\left(x+\frac{2\pi}{6}\right)-\tan\left(x+\frac{\pi}{6}\right)\right) \\ \qquad +\left(\tan\left(x+\frac{3\pi}{6}\right)-\tan\left(x+\frac{2\pi}{6}\right)\right)\end{array} +....+\left(\tan\left(x+\frac{n\pi}{6}\right)-\tan\left(x+(n-1)\frac{\pi}{6}\right)\right)\right]$$

$$\Rightarrow \quad f_1(x) = 2\left(\tan\left(x+\frac{n\pi}{6}\right)-\tan x\right)$$

$$\text{for } n=3,\ f_1(x) = 2\left(\tan\left(\frac{\pi}{2}+x\right)-\tan x\right) = 2(-\cot x-\tan x) = -2\frac{1}{\sin x\cos x}$$

$$2f_2(x) = f_1(x) - 2\tan\left(x+\frac{n\pi}{6}\right) = 2\tan\left(x+\frac{n\pi}{6}\right)-2\tan x-2\tan\left(x+\frac{n\pi}{6}\right)$$

$$\therefore \quad f_2(x) = -\tan x$$

$$f_3(x) = -f_2(x), \text{ so}$$

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$$\text{Now, } f_4(x) = \begin{cases} \frac{e^{(e^x-1)}-1}{e^{2(e^x-1)}} & ; x<0 \\ k_1 & ; x=0 \end{cases}$$

$$\text{Clearly, } f\left(0^-\right) = e^{\frac{1}{2}\ln e} = \sqrt{e}$$

$$f\left(0^-\right) = e^{k_2} \& f(0) = k_1$$

(i) $\text{As } f_4(x)$ **is continuous at** $x=0$**, so be definition of continuity, we get** $f\left(0^-\right) = f\left(0^+\right) = f(0)$

$$\sqrt{e} = e^{k_2} = k_1 \quad \therefore \quad k_1 = \sqrt{e} \& k_2 = \frac{1}{2}$$

(ii) As $y = f_3(x) = \tan x$

Clearly $f_3(x)$ **is continuous as well as derivable everywhere in** $(0, \pi/2)$

(iii) $n=3, f_1(x) = -4$

$$\Rightarrow \quad -2(\tan x+\cot x) = -4$$

$$\Rightarrow \quad \frac{\sin x}{\cos x}+\frac{\cos x}{\sin x} = 2$$

$$\Rightarrow \quad 1 = 2\sin x\cdot\cos x \quad \therefore \quad x = \frac{\pi}{4} \in \left(0, \frac{\pi}{2}\right)$$

Q.14 Answer:

$\pi/4$

Solution:

We have, $f_1(x) = 2\sum_{r=1}^{n} \frac{\sin\left(\left(x+\frac{r\pi}{6}\right)-\left(x+(r-1)\frac{\pi}{6}\right)\right)}{\left(x+(r-1)\frac{\pi}{6}\right)\cdot\cos\left(x+\frac{r\pi}{6}\right)}$

$=$

$$2\left[\begin{array}{l}\left(\tan\left(x+\frac{\pi}{6}\right)-\tan x\right)+\left(\tan\left(x+\frac{2\pi}{6}\right)-\tan\left(x+\frac{\pi}{6}\right)\right) \\ \qquad +\left(\tan\left(x+\frac{3\pi}{6}\right)-\tan\left(x+\frac{2\pi}{6}\right)\right)\end{array} +\ldots+\left(\tan\left(x+\frac{n\pi}{6}\right)-\tan\left(x+(n-1)\frac{\pi}{6}\right)\right)\right]$$

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$\Rightarrow \quad f_1(x) = 2\left(\tan\left(x + \frac{n\pi}{6}\right) - \tan x\right)$

for $n = 3, f_1(x) = 2\left(\tan\left(\frac{\pi}{2} + x\right) - \tan x\right) = 2(-\cot x - \tan x) = -2\frac{1}{\sin x \cos x}$

$2f_2(x) = f_1(x) - 2\tan\left(x + \frac{n\pi}{6}\right) = 2\tan\left(x + \frac{n\pi}{6}\right) - 2\tan x - 2\tan\left(x + \frac{n\pi}{6}\right)$

$\therefore \quad f_2(x) = -\tan x$

$f_3(x) = -f_2(x)$, so

$f_3(x) = \tan x$

$$\text{Now, } f_4(x) = \begin{cases} \frac{e^{(e^x-1)}-1}{e^{2(e^x-1)}} & ; x < 0 \\ k_1 & ; x = 0 \end{cases}$$

Clearly, $f\left(0^-\right) = e^{\frac{1}{2}\ln e} = \sqrt{e}$

$f\left(0^-\right) = e^{k_2} \& f(0) = k_1$

(i) As $f_4(x)$ is continuous at $x = 0$, so be definition of continuity, we get $f\left(0^-\right) = f\left(0^+\right) = f(0)$

$\sqrt{e} = e^{k_2} = k_1 \quad \therefore \quad k_1 = \sqrt{e}$ & $k_2 = \frac{1}{2}$

(ii) As $y = f_3(x) = \tan x$ Clearly $f_3(x)$ is continuous as well as derivable everywhere in $(0, \pi/2)$ (iii) $n = 3, f_1(x) = -4$

$\Rightarrow \quad -2(\tan x + \cot x) = -4$

$\Rightarrow \quad \frac{\sin x}{\cos x} + \frac{\cos x}{\sin x} = 2$

$\Rightarrow \quad 1 = 2\sin x \cdot \cos x \quad \therefore \quad x = \frac{\pi}{4} \in \left(0, \frac{\pi}{2}\right)$

Q.15 Answer:

0

Solution:

Since, the function is defined as three different polynomials in three different intervals, we have to test continuity and differentiability at the transition points 1 and 3 only.

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Continuity at x = 1,

$$f(1-) = \lim_{h\to 0} f(1-h) = \lim_{h\to 0} a(1-h)(1-h-1) + b = b$$

Hence for continuity at x = 1, b must be 0.

Continuity at x = 3, $f(3-0) = \lim_{h\to 0} (3-h-1) = 2$

$f(3) = 2$

$$f(3+0) = \lim_{h\to 0} f(3+h) = \lim_{h\to 0} p(3+h)^2 + q(3+h) + 2 = 9p + 3q + 2$$

Hence for continuity at x = 3, we must have

$9p + 3q + 2 = 2$ or $9p + 3q = 0$ ..... (1)

Derivative at x = 1,

$$f'(1-) = \lim_{h\to 0} \frac{f(1-h)-f(1)}{(1-h)-1} = \lim_{h\to 0} \frac{\{a(1-h)(-h)+0\}-0}{-h} = \lim_{h\to 0} a(1-h) = a$$

$$f'(1+) = \lim_{h\to 0} \frac{f(1+h)-f(1)}{(1+h)-1} = \lim_{h\to 0} \frac{(1+h-1)-0}{h} = 1$$

If $f'(1)$ is not to exist, a must not be equal to 1.

$$f'(3-) = \lim_{h\to 0} \frac{f(3-h)-f(3)}{3-h-3} = \lim_{h\to 0} \frac{(3-h-1)-2}{-h} = 1$$

$$f'(3+) = \lim_{h\to 0} \frac{f(3+h)-f(3)}{3+h-3} = \lim_{h\to 0} \frac{\{p(3+h)^2+q(3+h)+2\}-2}{h}$$

$= 6p + q$, **since** $9p + 3q = 0$ **in view of eq. (1)**

For differentiability at $x = 3, 6p + q = 1$

Solve eq. (1) and (3), we get $p = \frac{1}{3}, q = -1$

Hence, $a \neq 1, b = 0, p = \frac{1}{3}, q = -1$

Q.16 Answer:

$1/3$

Solution:

Since, the function is defined as three different polynomials in three different intervals, we have to test continuity and differentiability at the transition points 1 and 3 only.

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Continuity at x = 1,

$$f(1-) = \lim_{h\to 0} f(1-h) = \lim_{h\to 0} a(1-h)(1-h-1) + b = b$$

Hence for continuity at x = 1, b must be 0.

Continuity at x = 3, $f(3-0) = \lim_{h\to 0} (3-h-1) = 2$

$f(3) = 2$

$$f(3+0) = \lim_{h\to 0} f(3+h) = \lim_{h\to 0} p(3+h)^2 + q(3+h) + 2 = 9p + 3q + 2$$

Hence for continuity at x = 3, we must have

$9p + 3q + 2 = 2$ or $9p + 3q = 0$ ..... (1)

Derivative at x = 1,

$$f'(1-) = \lim_{h\to 0} \frac{f(1-h)-f(1)}{(1-h)-1} = \lim_{h\to 0} \frac{\{a(1-h)(-h)+0\}-0}{-h} = \lim_{h\to 0} a(1-h) = a$$

$$f'(1+) = \lim_{h\to 0} \frac{f(1+h)-f(1)}{(1+h)-1} = \lim_{h\to 0} \frac{(1+h-1)-0}{h} = 1$$

If $f'(1)$ is not to exist, a must not be equal to 1.

$$f'(3-) = \lim_{h\to 0} \frac{f(3-h)-f(3)}{3-h-3} = \lim_{h\to 0} \frac{(3-h-1)-2}{-h} = 1$$

$$f'(3+) = \lim_{h\to 0} \frac{f(3+h)-f(3)}{3+h-3} = \lim_{h\to 0} \frac{\{p(3+h)^2+q(3+h)+2\}-2}{h}$$

$= 6p + q$, **since** $9p + 3q = 0$ **in view of eq. (1)**

For differentiability at $x = 3, 6p + q = 1$

Solve eq. (1) and (3), we get $p = \frac{1}{3}, q = -1$

Hence, $a \neq 1, b = 0, p = \frac{1}{3}, q = -1$

Q.17 Answer:

-1

Solution:

Since, the function is defined as three different polynomials in three different intervals, we have to test continuity and differentiability at the transition points 1 and 3 only.

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Continuity at x = 1,

$$f(1-) = \lim_{h\to 0} f(1-h) = \lim_{h\to 0} a(1-h)(1-h-1) + b = b$$

Hence for continuity at x = 1, b must be 0.

Continuity at x = 3, $f(3-0) = \lim_{h\to 0} (3-h-1) = 2$

$f(3) = 2$

$$f(3+0) = \lim_{h\to 0} f(3+h) = \lim_{h\to 0} p(3+h)^2 + q(3+h) + 2 = 9p + 3q + 2$$

Hence for continuity at x = 3, we must have

$9p + 3q + 2 = 2$ or $9p + 3q = 0$ ..... (1)

Derivative at x = 1,

$$f'(1-) = \lim_{h\to 0} \frac{f(1-h)-f(1)}{(1-h)-1} = \lim_{h\to 0} \frac{\{a(1-h)(-h)+0\}-0}{-h} = \lim_{h\to 0} a(1-h) = a$$

$$f'(1+) = \lim_{h\to 0} \frac{f(1+h)-f(1)}{(1+h)-1} = \lim_{h\to 0} \frac{(1+h-1)-0}{h} = 1$$

If f ' (1) is not to exist, a must not be equal to 1.

$$f'(3-) = \lim_{h\to 0} \frac{f(3-h)-f(3)}{3-h-3} = \lim_{h\to 0} \frac{(3-h-1)-2}{-h} = 1$$

$$f'(3+) = \lim_{h\to 0} \frac{f(3+h)-f(3)}{3+h-3} = \lim_{h\to 0} \frac{\{p(3+h)^2+q(3+h)+2\}-2}{h}$$

$= 6p + q$, **since** $9p + 3q = 0$ **in view of eq. (1)**

For differentiability at $x = 3, 6p + q = 1$

Solve eq. (1) and (3), we get $p = \frac{1}{3}, q = -1$

Hence, $a \neq 1, b = 0, p = \frac{1}{3}, q = -1$

Q.18 Answer:

A-p ; B-p,q ; C-p,r

Solution:

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(A) - p, (B) - pq, (C) - pr

$$(\mathrm{A})\ \text{R.H.L.} = \lim_{h\to 0} \frac{\left(\frac{\pi}{2} - \sin^{-1}\left(1-h^2\right)\right)\sin^{-1}(1-h)}{\sqrt{2}\,(h-h^3)}$$

$$= \lim_{h\to 0} \frac{\cos^{-1}\left(1-h^2\right)\sin^{-1}(1-h)}{\sqrt{2}h(1-(h))^2} = \lim_{h\to 0} \frac{\sin^{-1}\sqrt{2h^2-h^2}\sin^{-1}(1-h)}{\sqrt{2}h(1-(h))^2}$$

$$= \lim_{h\to 0} \frac{\sin^{-1}\left(h\sqrt{2-h^2}\right)\sin^{-1}(1-h)}{\sqrt{2}h(1-(h))^2} = \frac{\pi}{2} \Rightarrow \mathrm{k} = \frac{\pi}{2}$$

$$\text{L.H.L.} = \lim_{h\to 0} A\frac{\sin^{-1}(1-(1-h))\cos^{-1}(1-(1-h))}{\sqrt{2(1-h)}\cdot h} = \lim_{h\to 0} A\frac{\sin^{-1}h\cos -1}{\sqrt{2(1-h)}\cdot h} = A\frac{\pi}{2\sqrt{2}}$$

$$\Rightarrow\ \mathrm{A}\ \frac{\pi}{2\sqrt{2}} = \frac{\pi}{2} \Rightarrow A = \sqrt{2}$$

Hence, $\sin^2\mathrm{k} + \cos^2\left(\frac{\mathrm{A}\pi}{\sqrt{2}}\right) = 1+1 = 2$

(B) We have $[2 + 5|\mathrm{n}|\sin\mathrm{x}] = 2 + [5|\mathrm{n}|\sin\mathrm{x}]$

Let $\mathrm{y} = 5|\mathrm{n}|\sin\mathrm{x}$

$\therefore$ **Number of points of non-derivability**

$= 2(5|\mathrm{n}| - 1) + 1 = 10|\mathrm{n}| - 1$

$\therefore\quad 10|\mathrm{n}| - 1 = 19 \Rightarrow 10|\mathrm{n}| = 20 \Rightarrow |\mathrm{n}| = 2$

Hence, $\mathrm{n} = \pm 2$

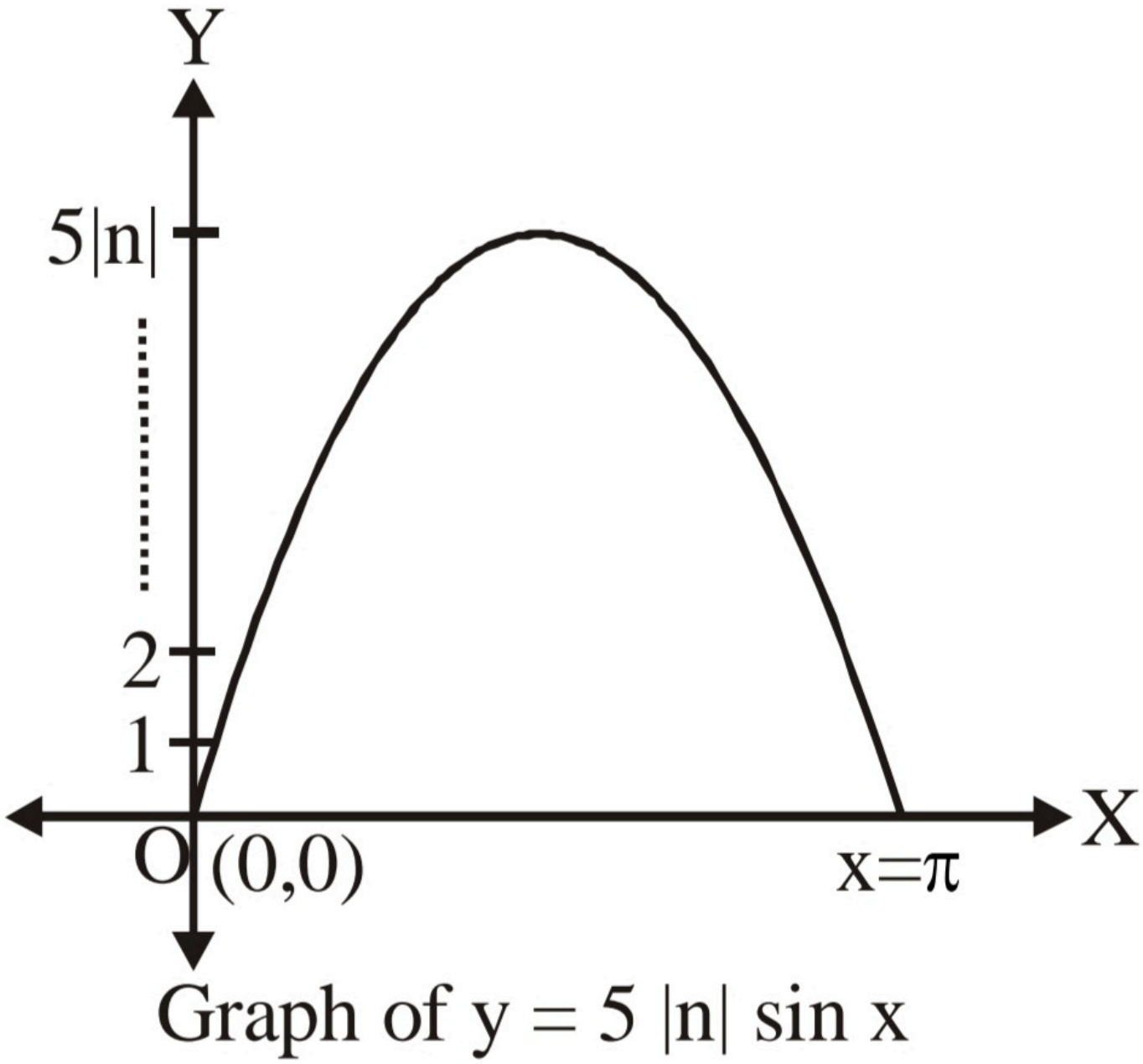
(C) R.H.D. $= \lim_{h\to 0} h\frac{\frac{(3/4)^{1/h} - (3/4)^{-1/h}}{(3/4)^{1/h} + (3/4)^{-1/h}}}{h} = -1$

L.H.D. $= \lim_{h\to 0}(-h)\frac{\frac{(3/4)^{-1/h} - (3/4)^{1/h}}{(3/4)^{-1/h} - (3/4)^{1/h}}}{-h} = 1$

$\therefore\quad P = f'(0^-) - f'(0^+) = 2$

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Now, $\lim_{x \to 2^-} \frac{(\exp((x+2)\ln 4))^{\frac{\lfloor x+1 \rfloor}{4}} - 16}{4^x - 16} = \frac{1}{2}$



Graph of y = 5 |n| sin x

Q.19 Answer:

—

Solution:

$OB_k = k$

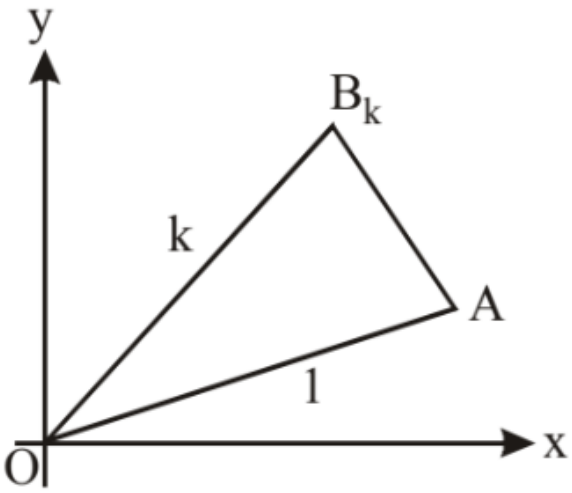
$\angle AOB_k = \frac{k\pi}{2n}$

$S_k = \frac{1}{2} k \sin \frac{k\pi}{2n} \left(\text{Using} \Delta = \frac{1}{2} ab \sin\theta\right)$

$L = \frac{k}{2n^2} \sum_{n=1}^{\infty} \sin \frac{k\pi}{2n} = \frac{1}{2n} \sum_{n=1}^{\infty} \sin \frac{k\pi}{2n} = \frac{1}{2} \int_0^1 x \cdot \sin \frac{\pi x}{2} dx$

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$= \frac{1}{2}[\underbrace{-\frac{2}{\pi}x\cos\frac{\pi x}{2}\Big|_0^1}_{\text{zero}} + \frac{2}{\pi}\int_0^1 \cos\frac{\pi x}{2}dx] = \frac{1}{2}\left[0 + \frac{2}{\pi}\cdot\frac{2}{\pi}\left(\sin\frac{\pi x}{2}\right)_0^1\right] = \frac{2}{\pi^2}$



Q.20 Answer:

—

Solution:

For limit to exist and equal to 1 coefficient of x^4 **in denominator** $= 0$ **(as degree of** x **in** $D^r > N^r$**)**

Now degree of x **in** D^r **is 2 and degree of** x **in** N^r **is 3**

$\therefore$ **coefficient of** x^3 **in** $N^r = 0$ **otherwise** $L \neq 1$ **and will be infinity and** $\frac{\text{coefficient of } x^2 \text{ in } N^r}{\text{coefficient of } x^2 \text{ in } D^r} = 1$ **now coefficient of** x^4 **in** $D^r = 5a - b + 4c = 0$

coefficient of x^3 **in** $N^r = 2a + b - 3c = 0$

$\frac{\text{coefficiently of } x^2 \text{ in } N^r}{\text{coefficiently of } x^2 \text{ in } D^r} = 1 \Rightarrow \frac{-a+5b-c}{2} = 1$

$\therefore \quad a - 5b + c + 2 = 0$

Solving (1), (2) and (3) we get a $= \frac{-2}{109}, b = \frac{46}{109}$ **and** $c = \frac{14}{109}$

$\Rightarrow \quad a + b + c = \frac{58}{109} = \frac{p}{q} \Rightarrow p + q = 167$

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